

Session Handoff & Context

Current Phase: Characterization & Audio Finalized (v5.1.4)

We successfully executed the **v5.0.0 Unified nanoTesla (nT) Architecture** and resolved a series of cascading edge-cases that emerged when pushing the hardware to its absolute limits.

The firmware now sits at **v5.1.4**, boasting the following achievements:

1. **Saturation Physics Discovered:** We discovered that the RM3100 experiences two distinct failure modes: Physical Core Collapse (at $\sim 533 \mu\text{T}$, regardless of cycle count) and Digital Integer Overflow (e.g. at 3200 CC, maxing out around $88 \mu\text{T}$). This physics realization heavily anchors our understanding of the sensor's absolute dynamic range.
2. **Logarithmic / Cube-Root Audio:** The audio feedback algorithm was fundamentally rewritten to use a mathematically pure `cbrtf()` (cube root) algorithm. Because magnetic dipole fields decay at $1/r^3$, this cube root dynamically maps the audio pitch linearly to the physical $1/r$ distance of the anomaly. The UI Gain knob now dynamically scales the maximum expected physical range boundary (e.g. 1000 nT to 500,000 nT) rather than applying a raw multiplier.
3. **The nT Consistency Bug:** Eradicated. `SensorFusion` now natively ingests raw counts, immediately converts them to physical nT using the 0.38 scalar, and applies a purely physical calibration matrix. The calculated gradient magnitude remains flawlessly consistent regardless of Cycle Count jumps.

Where We Left Off

All architecture documentation (`README.md`, `project_overview.md`, `user_manual.md`) has been brought up to date to reflect the v5.1.4 state, including the addition of the new `MFS_Wiring_Diagram.png` schematic. The codebase is clean, committed, and stable. The Python scripts (`characterize_system.py`) were also updated to intelligently find the true saturation plateaus using histogram modes.

Next Steps for the Next Agent

Wait for the user's direction. We are ready to tackle new functional features (like the 3rd sensor or GPS integration) or refine any remaining UI quirks. The user was last discussing improving the physical characterization protocols using a threaded rod for precise $1/r$ measurements.

Note to Agent:

When powering on the ESP32 via the physical battery button, the PMIC requires the ESP32 to assert the `sys_EN` pin HIGH via the TCA9554 IO Expander to latch power. Do not move I2C init below `Serial` waits.